Title of the Project:

Inventory & Billing System (Retail Shop App) – Spring Boot & React Based

Problem Statement:

Retail shops often struggle with inaccurate stock tracking, delayed billing, manual invoice preparation, and lack of real-time sales insights.
This results in financial losses, stockouts, human errors, and inefficient daily operations.
A centralized automated system is required to manage inventory, generate bills instantly, maintain sales records, and alert the shop owner about low-stock items.

Objectives & Scope:

- Automate the billing and inventory process of a retail store.
- Maintain stock levels in real-time.
- Generate professional PDF invoices instantly.
- Provide an admin dashboard with analytics and sales reports.
- Enable low-stock alerts to prevent product shortages.
- Ensure secure role-based access for admin and staff.

Process Description:

The system includes modules for Billing, Inventory Management, User Management, and Admin Reporting.
Spring Boot handles backend APIs, authentication, and database operations.
React provides a user-friendly interface for shop staff.
The data flow includes adding products, updating stock, processing bills, and generating invoices.
The system maintains logs and updates stock after every sale. It provides a dashboard showing sales trends, revenue, and low-stock alerts.

Hardware & Software Requirements:

Hardware: Standard PC/Laptop, 4–8 GB RAM

Software:

- Backend: Java, Spring Boot, Spring Data JPA
- Frontend: ReactJS
- Database: MySQL
- Tools: IntelliJ IDEA, VS Code, Postman
- Libraries: Jasper, Reports/ OpenPDF

Future Work:

- Integration with barcode scanner.
- GST-compliant invoice formats.
- Mobile app version using Flutter.
- Cloud deployment on AWS.
- Multi-branch shop support.

Limitations:

- No hardware integration like barcode scanners or thermal printers in the basic version.
- Suitable mainly for small shops and not optimized for large enterprises.
- No mobile application or offline billing support available.
- Product entry is manual and time-consuming during the initial setup.
- Supports only a single-branch operation

Project Schedule (Gantt Representation):

Week 1–2: Requirement analysis and system design

Week 3–4: Spring Boot backend setup and database design

Week 5–6: Frontend development using React

Week 7: Billing module and PDF invoice generation

Week 8: Dashboard and analytics implementation

Week 9: Testing and debugging

Week 10: Final integration and documentation

References:

[1] Craig Walls, *Spring in Action*, Manning Publications, 2018.

[2] Eric Freeman, *Head First Design Patterns*, O'Reilly Media, 2020.

[3] ReactJS Documentation, https://react.dev

[4] MySQL Documentation, https://dev.mysql.com